



The Solar System — FULL MIND MAP

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Chapter 01_ii_The_Solar_System

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Solar System — Composition

- Solar System ~4.6 billion years pehle bana; isme Sun, planets, dwarf planets, satellites, asteroids, comets, meteoroids, cosmic dust hote hain. [Q1]
- **Nebula** Solar System ka part NAHI hai (baaki sab options — asteroids, comets, planets — part hain). [Q1]

Heliocentric Model — Copernicus

- **Nicolaus Copernicus** ne modern heliocentric model diya — Earth Sun ki parikrama karti hai, na ki centre mein hai (16th century, Scientific Revolution ki shuruaat). [Q2, Q3, Q4, Q5]
- Baad mein **Galileo** ke observations aur **Kepler** ke orbital laws ne heliocentrism ko support diya. [Q2, Q3, Q4]

Planets — Count aur Order

- Currently **8 planets** recognized hain: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune (2006 mein Pluto ko dwarf planet declare kiya gaya). [Q7]
- Sun se badhti distance order: Mercury → Venus → Earth → Mars → Jupiter → Saturn → Uranus → Neptune. [Q8, Q9, Q10, Q11]
- Mars aur Uranus ke beech: **Jupiter aur Saturn** hain. [Q8]
- Earth **Venus aur Mars** ke beech located hai; **Mercury aur Venus** ke beech koi planet nahi (Earth Sun-Venus ke beech nahi). [Q10, Q11]



EXAM TRAP

- Q10 mein 'Earth-Sun ke beech planets' poocha — answer **Mercury aur Venus** (dono Earth ki orbit se andar hain), Mars/Venus wala combo Q11 (Earth ke around) se confuse mat karo. [Q10, Q11]

Planet Sizes — Largest/Smallest, Order

- Size order (descending): **Jupiter > Saturn > Uranus > Neptune > Earth > Venus > Mars > Mercury.** [Q13,Q16]
- **Mercury** = smallest planet; **Jupiter** = largest planet. [Q13,Q14]
- **Neptune** = slowest orbital-velocity planet (NOT Mars) — ~165 years mein 1 revolution. [Q15]

Planet Composition, Density, Light

- Earth ka average density sabse zyada (~5.5 g/cm³) — Mercury 5.4, Venus 5.2, Mars 3.9, Jupiter 1.3, Saturn 0.7 (lowest), Uranus 1.3, Neptune 1.6. [Q6]
- Earth composition: Iron (32%), Oxygen (30%), Silicon (15%) — silicon predominant NAHI hai. [Q6]
- Sun ka mass Solar System ke total mass ka ~99.8% hai (75% nahi); Sun ka diameter Earth ka ~109.2 times hai (190 times nahi). [Q6]
- Jupiter aur Saturn mein mainly **Hydrogen aur Helium** hoti hai, saath mein Methane bhi. [Q17]
- Planets **non-luminous** bodies hain (khud light produce nahi karte) lekin Sun ki light reflect karke chamakte hain. [Q12]

Match/Identification Questions

- Smallest planet = Mercury; Largest = Jupiter; 2nd planet from Sun = Venus; Farthest recognized planet = Neptune. [Q13]
- **Florida** USA ka state hai, planet nahi (Mercury, Venus, Saturn planets hain). [Q18]
- Uranus = planet, Moon = natural satellite, Halley = comet, Mariner = artificial satellite/spacecraft. [Q20]

Goldilocks Zone aur Solar System Facts

- **Goldilocks Zone** (circumstellar habitable zone) = star ke around wo region jahan liquid water exist kar sake — Earth is zone mein hai, isliye life-bearing hai. [Q19]
- Solar System ka chief celestial body **Sun** hai (centre mein located). [Q22]
- Venus Earth ke sabse close aa sakta hai (closest-approach distance ke hisaab se). [Q11]

⚠ EXAM TRAP

- Statements set: Mercury sabse hot planet NAHI (Venus hai); Ganymede Saturn ka NAHI, Jupiter ka moon hai; Neptune ke rings methane gas ke NAHI, dust/ice particles ke hain; sirf Phobos-Deimos (Mars ke moons) wala statement correct hai. [Q21]